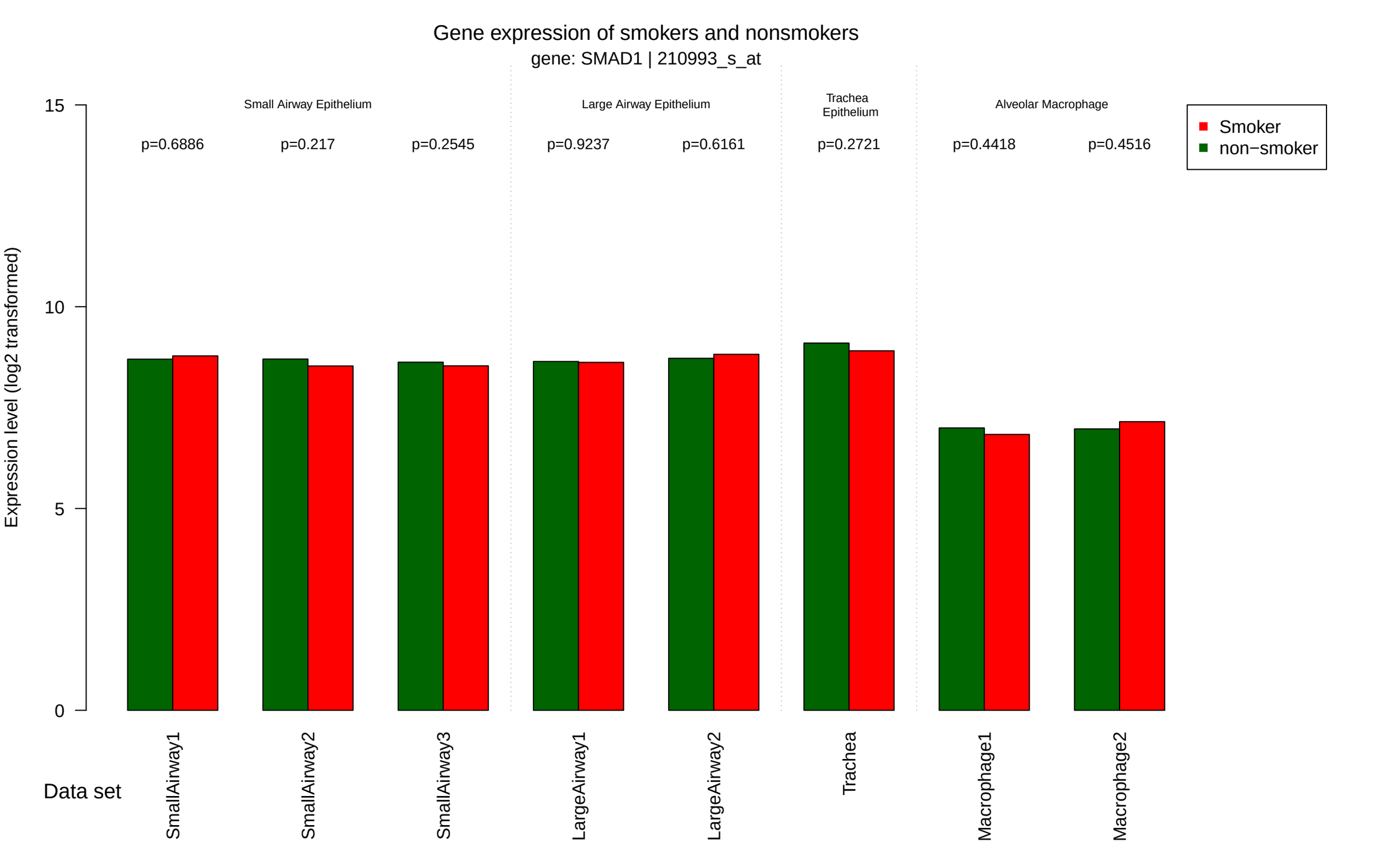
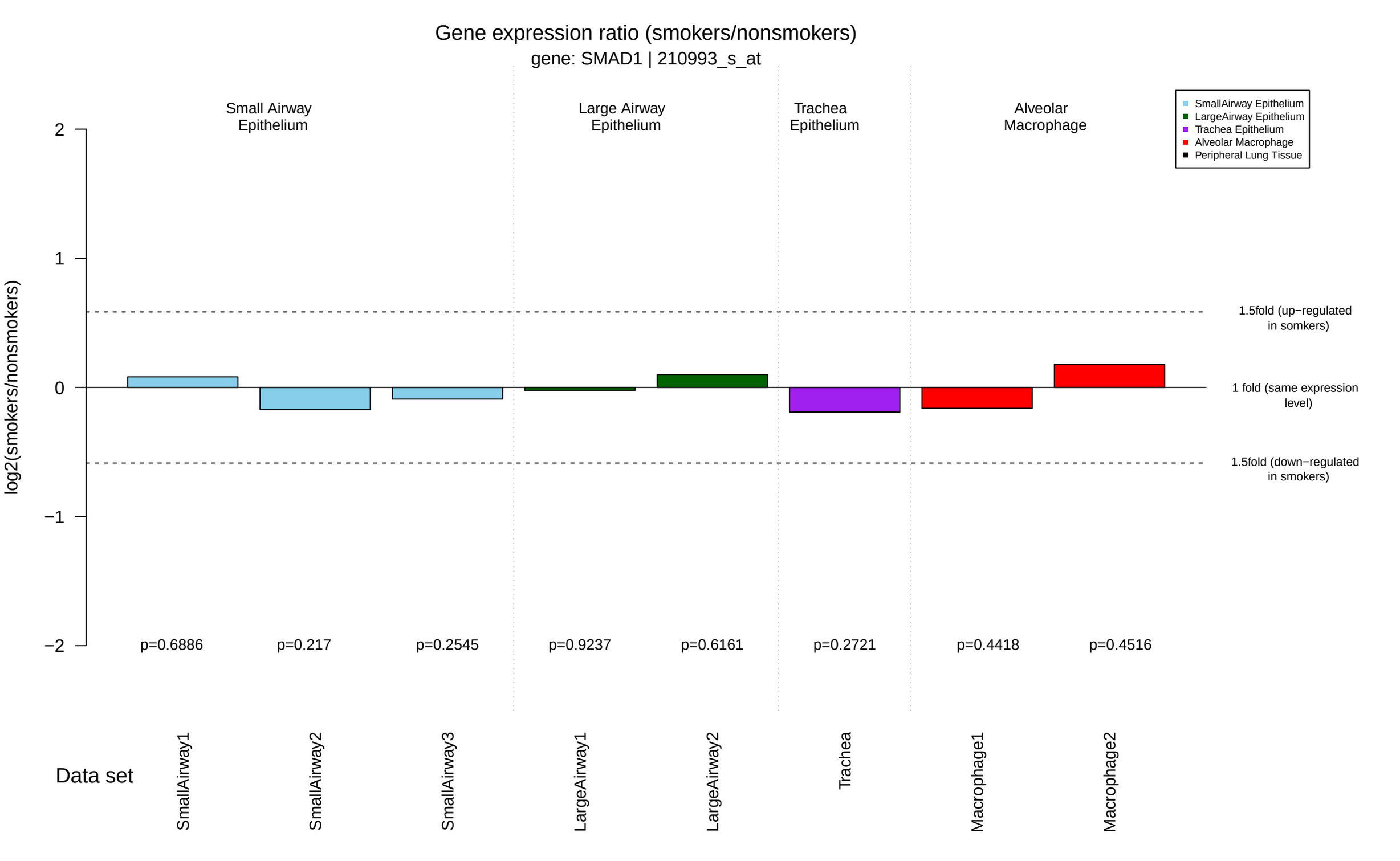
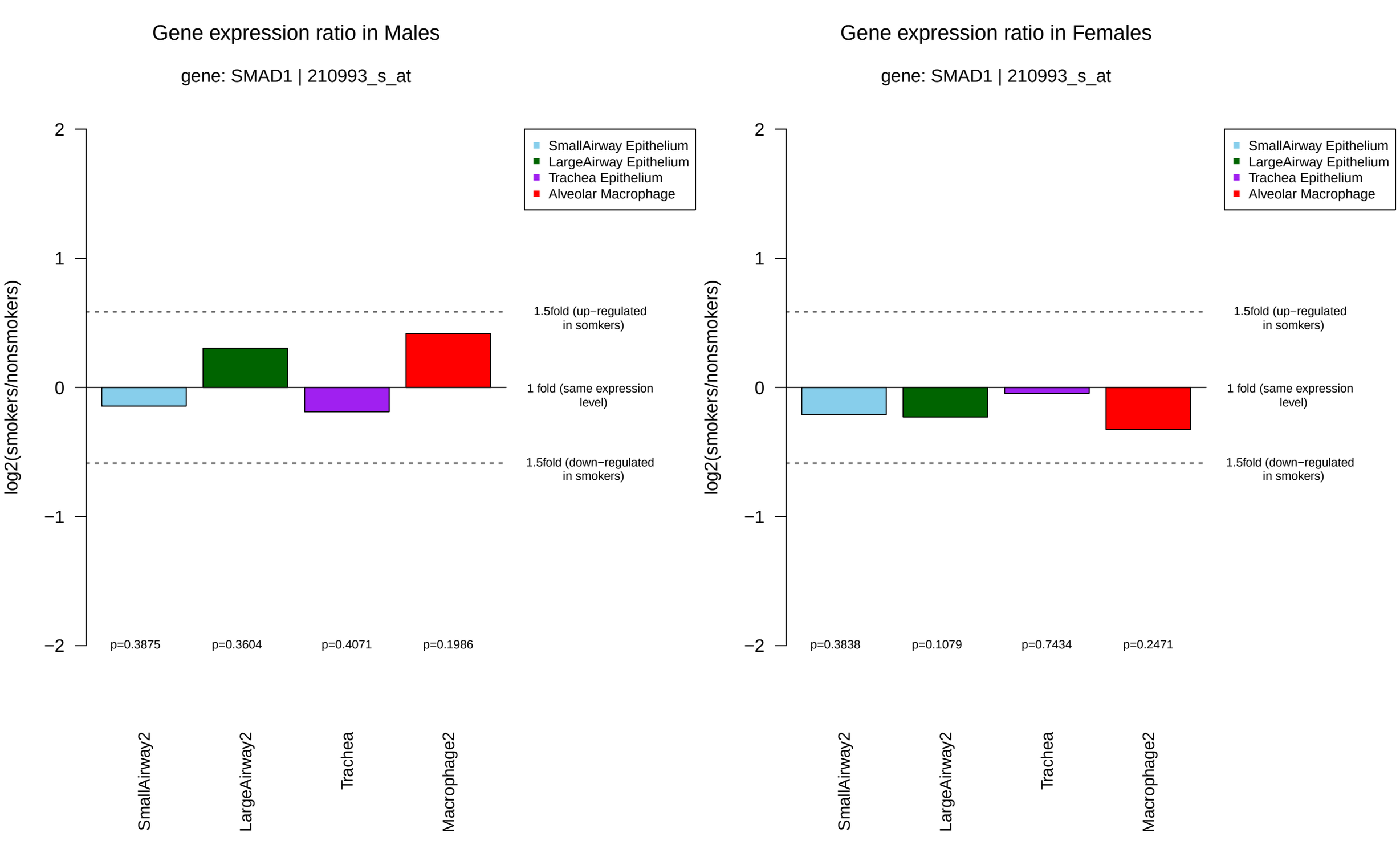
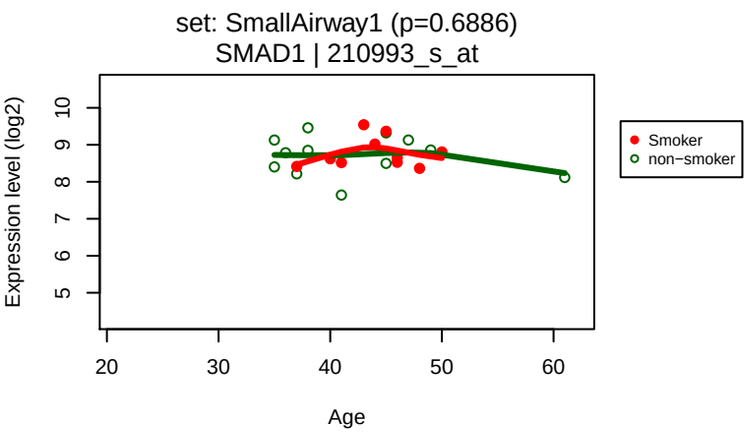
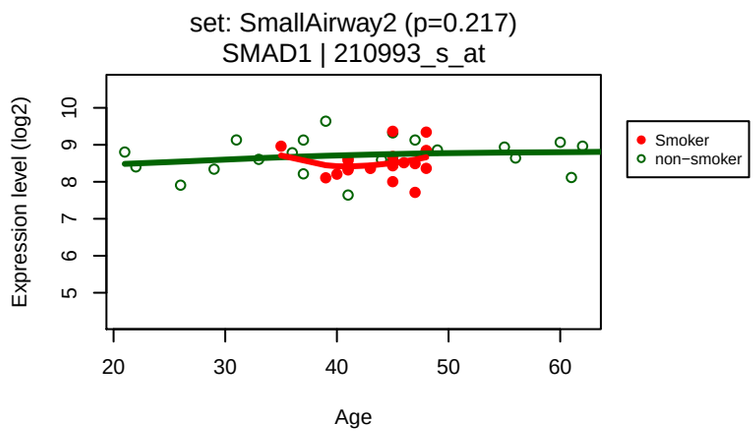
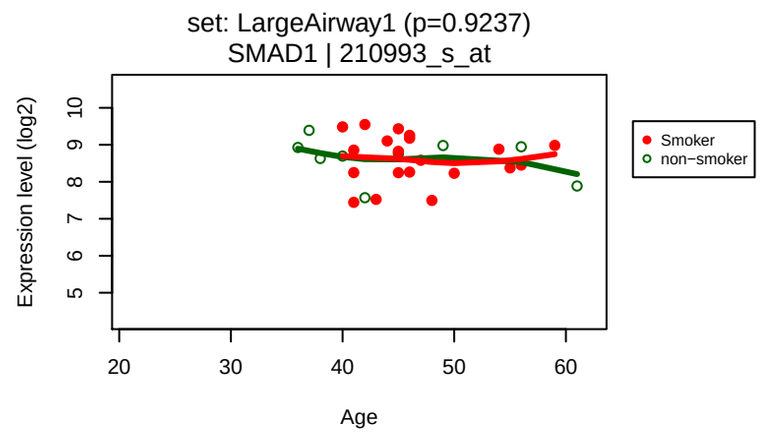

Gene expression of smokers and nonsmokers
gene: SMAD1 | 210993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6886
p=0.217
p=0.2545
p=0.9237
p=0.6161
p=0.2721
p=0.4418
p=0.4516
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

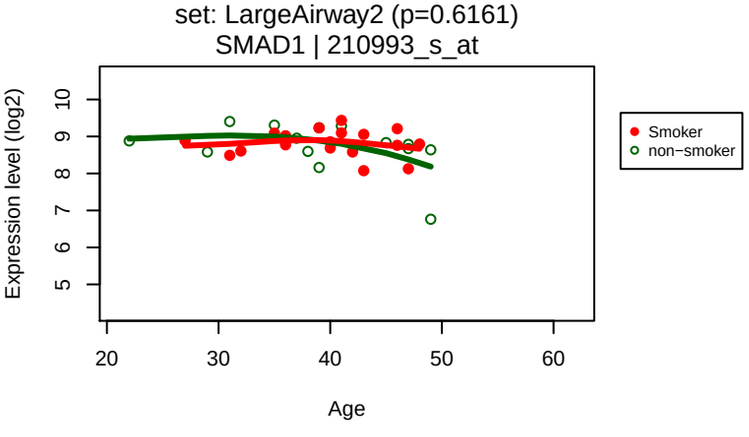

Gene expression ratio (smokers/nonsmokers)
gene: SMAD1 | 210993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6886
p=0.217
p=0.2545
p=0.9237
p=0.6161
p=0.2721
p=0.4418
p=0.4516
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

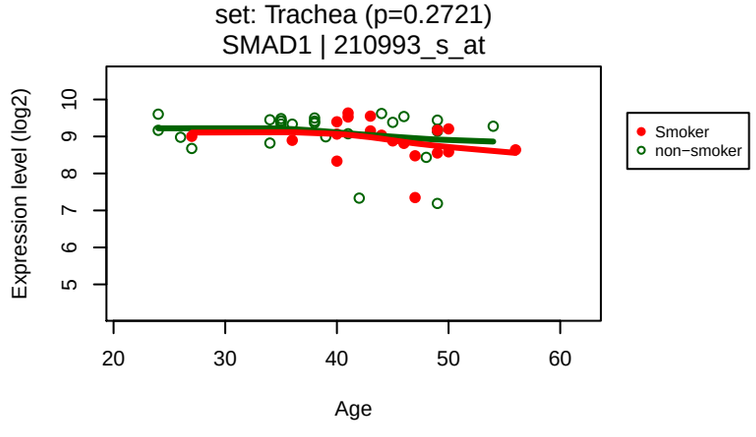

Gene expression ratio in Males
gene: SMAD1 | 210993_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3875
p=0.3604
p=0.4071
p=0.1986
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD1 | 210993_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3838
p=0.1079
p=0.7434
p=0.2471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

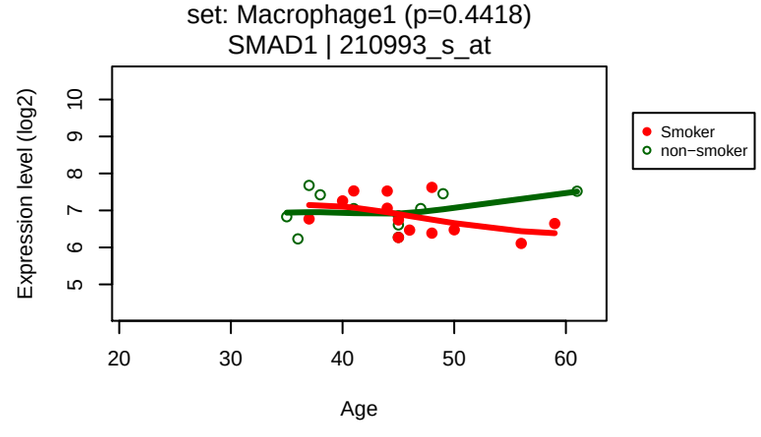

set: SmallAirway1 (p=0.6886)
SMAD1 | 210993_s_at
Expression level (log2)
Age
Smoker
non−smoker

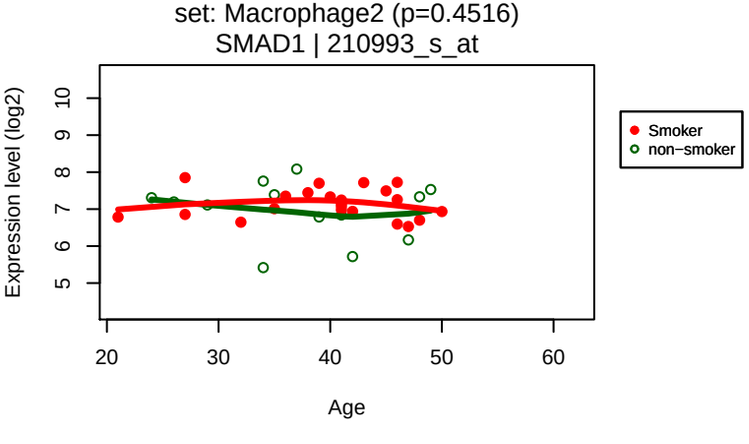

set: SmallAirway2 (p=0.217)
SMAD1 | 210993_s_at
Expression level (log2)
Age
Smoker
non−smoker

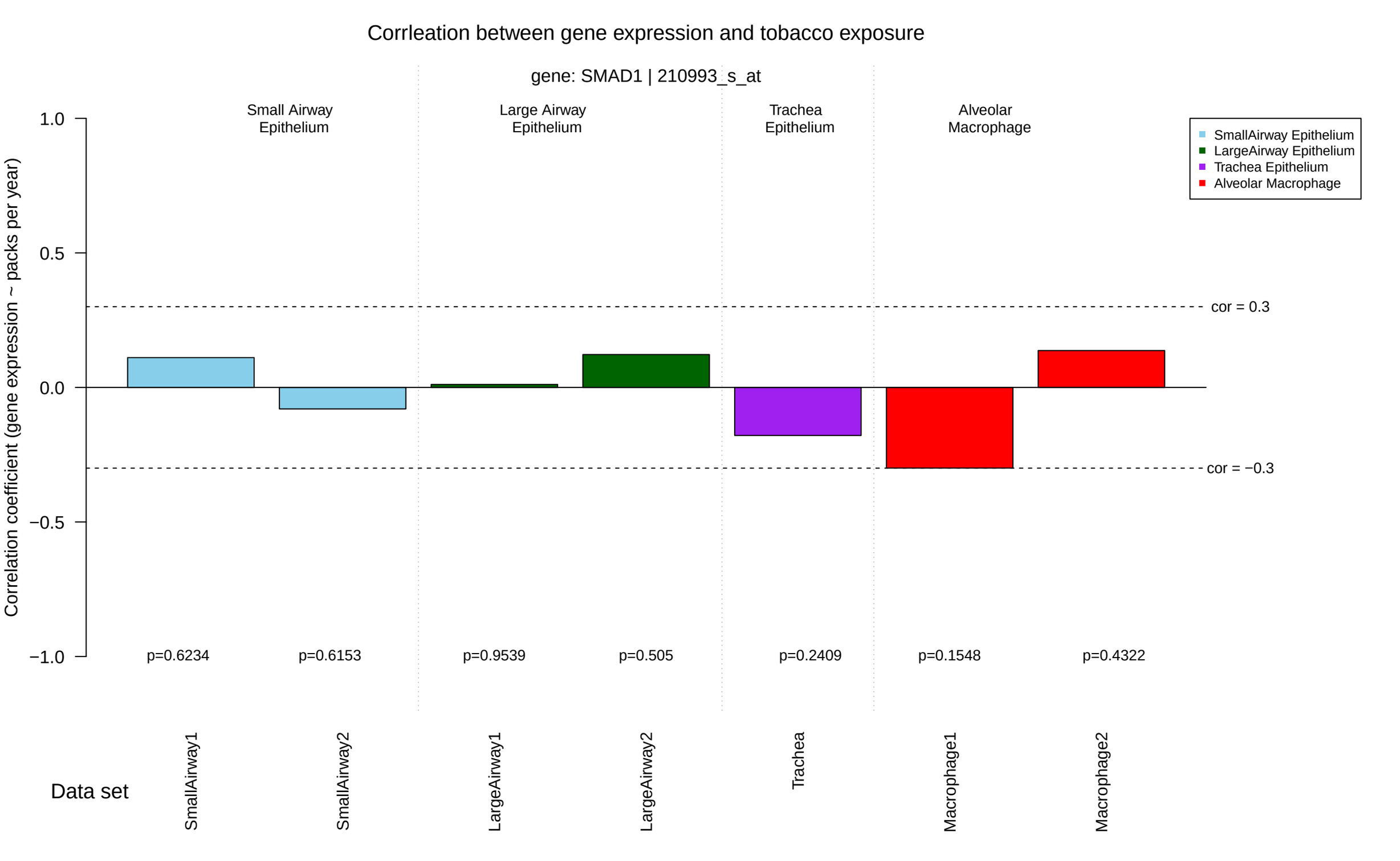

set: LargeAirway1 (p=0.9237)
SMAD1 | 210993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6161)
SMAD1 | 210993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2721)
SMAD1 | 210993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4418)
SMAD1 | 210993_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4516)
SMAD1 | 210993_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD1 | 210993_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6234
p=0.6153
p=0.9539
p=0.505
p=0.2409
p=0.1548
p=0.4322
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2